



DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

B23CBT501 - FULL STACK DEVELOPMENT

QUESTION BANK -2

Q.No.	Question	CO
1.	What is a NoSQL database? a). A relational database b). A non-relational database c). A database for SQL queries d). A database for storing images	CO3
2.	Which data format does MongoDB use to store data? a). XML b). JSON c). CSV d). BSON	CO3
3.	What is the MongoDB command to show all databases? a). show dbs b). list databases c). display databases d). show all	CO3
4.	Which of the following is used for authentication in MongoDB? a). User authentication b). IP whitelisting c). TLS/SSL encryption d). A and C	CO3
5.	What is Express in Node.js? a). A relational database b). A web application framework c). A database driver d). A programming language	CO4
6.	Which Express middleware is used to parse JSON data in the request body? a). express.json() b). express.urlencoded() c). bodyParser() d). express.body()	CO4
7.	What does the next() function do in Express middleware? a). Ends the request-response cycle b). Passes the request to the next middleware c). Initializes the response d). Starts the server	CO4
8.	Which of the following is the primary language used in Angular? a). JavaScript b). TypeScript c). Python d). Ruby	CO4
9.	Which MongoDB feature helps distribute data across multiple servers? a). Replication b). Sharding c). Indexing d). Aggregation	CO3
10.	What is the main purpose of MongoDB indexes? a). To speed up queries b). To delete data quickly d). To store metadata d). To encrypt data	CO3

11.	Which MongoDB method is used to count the number of documents in a collection? a). db.collection.count() b). db.collection.size() c). db.collection.getCount() d). db.collection.numDocs()	CO3
12.	Which of the following is an advantage of MongoDB over traditional relational databases? a). It supports SQL queries b). It provides ACID compliance c). It is highly scalable and flexible d). It supports transactions across multiple databases	CO3
13.	In Angular, what is a Component? a). A directive b). A reusable UI element c). A controller for business logic d). A module for routing	CO4
14.	What is the decorator used to define an Angular component? a). @Directive b). @Injectable c). @Component d). @NgModule	CO4
15.	What is the command to serve an Angular application in development mode? a). ng build b). ng serve c). ng run d). ng start	CO4
16.	What is an Angular module? a). A component of the application b). A group of components, services, and other code c). A class to handle HTTP requests d). A file for routing configuration	CO4
Part B		2 Marks
Q.No.	Question	CO
1.	What is the purpose of the mongod command?	CO3
2.	Name any two roles that can be assigned to a user in MongoDB.	CO3
3.	Recall the command to create a collection in MongoDB?	CO3
4.	List the built-in roles available in MongoDB.	CO3
5.	Label the advantages of using TypeScript with Angular.	CO4
6.	How does Express.js handle multiple routes for different HTTP methods?	CO4
7.	Compare static and dynamic routing in Express.	CO4
8.	Why is Express.js preferred over the default HTTP module in Node.js?	CO4
9.	Name the default port number for MongoDB?	CO3
10.	Recall the function of the MongoDB configuration file?	CO3
11.	What is the command to drop a database in MongoDB?	CO3
12.	List the significances of authentication in MongoDB?	CO3
13.	Compare JavaScript and TypeScript in terms of features.	CO4
14.	How does Express.js simplify the creation of REST APIs?	CO4
15.	Why is static typing beneficial in Angular applications?	CO4
16.	Label the different types of data binding supported in Angular.	CO4

Part C		13 MARKS	
Q. No.	Question	Marks	CO
1.a)	Illustrate a web-based form application using Node.js and MongoDB that accepts user input and stores it in a database. Explain the flow from form submission to database storage.	(13)	CO3
b)	Demonstrate the Role-Based Access Control (RBAC) in MongoDB? Describe how to create roles and assign them to users with appropriate examples.	(13)	CO3
2.a)	Apply property binding and event binding in a simple Angular component to build a like/dislike counter. Show the working HTML and TypeScript code.	(13)	CO4
b)	Develop an Angular form using template-driven approach. Include form validation and display appropriate messages for each field.	(13)	CO4
3.a)	Illustrate the various database administration tasks in MongoDB such as creating, switching, listing, and dropping databases. Provide relevant commands.	(13)	CO3
b)	Demonstrate a simple Node.js application that uses Express.js and MongoDB to perform Create, Read, Update, and Delete (CRUD) operations on a 'students' collection.	(13)	CO3
4.a)	Develop a basic Express.js server that handles different HTTP methods (GET, POST, PUT, DELETE) for a /users route. Explain how req and res are used in each route.	(13)	CO4
b)	Construct a contact form backend using Express where the frontend sends user details (name, email, message). Handle POST requests, validate data (non-empty fields) and return success or error messages.	(13)	CO4